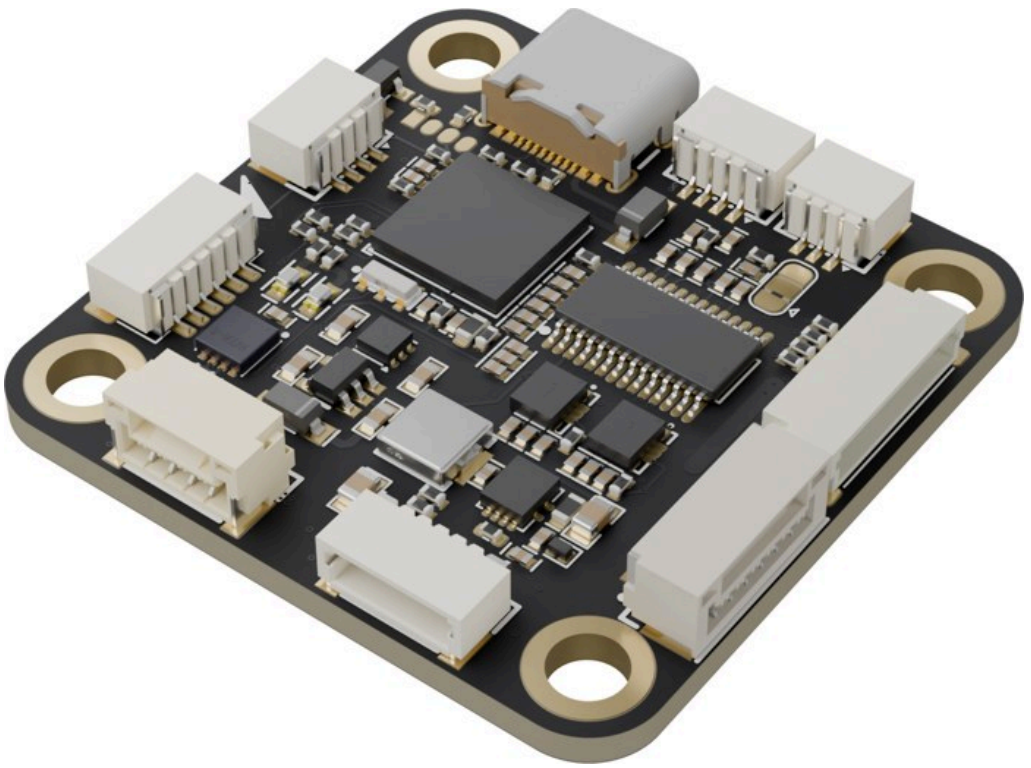
AF-H7 nano V2

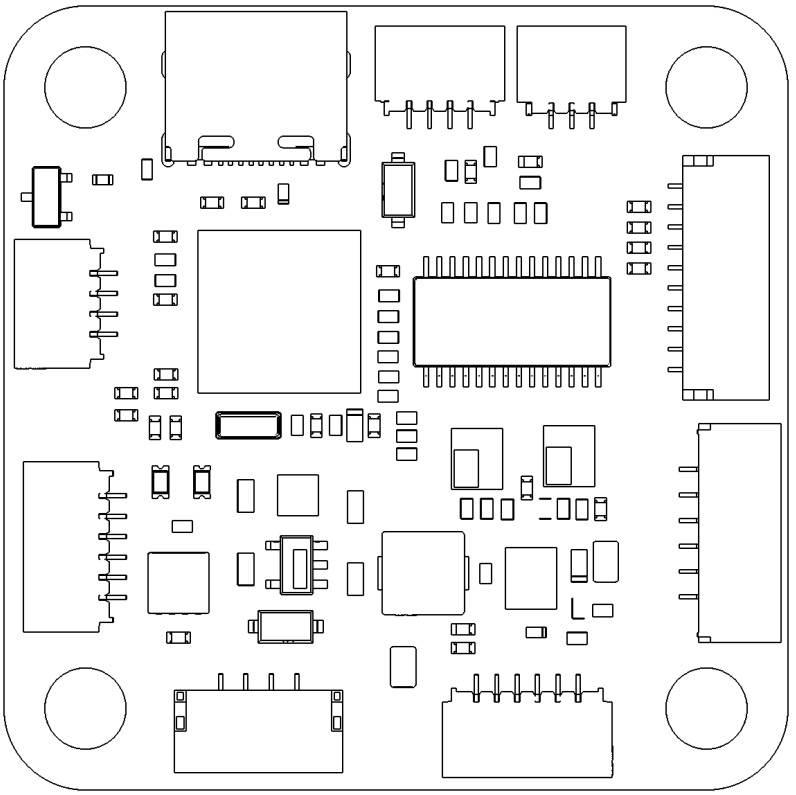
User Guide

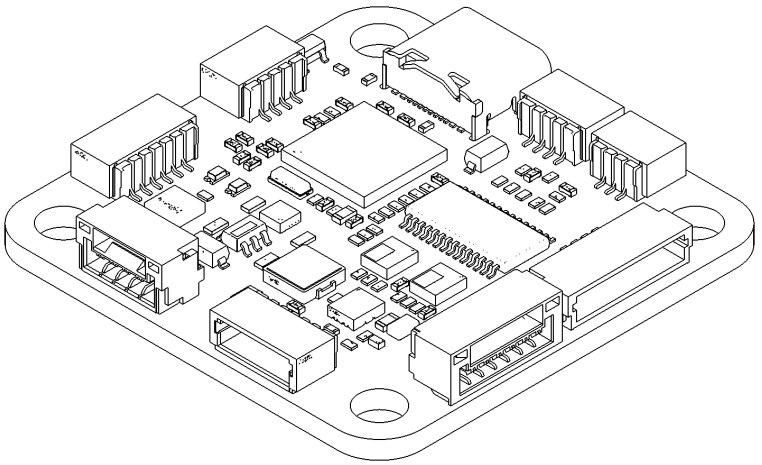


novaX

1. Parts List

· Main Board





AF-H7 nano V2 * 1

1. Parts List

· Connectors & Pads

Port	Type
ESC	JST-SH 1.0 mm 10P (JST SM10B-SRSS-TB)
OpenIPC	JST-GH 1.25 mm 6P
SWD	4 solder pads
RC	JST-SH 1.0 mm 4P
GPS	JST-SH 1.0 mm 6P
VTX	JST-SH 1.0 mm 4P
CAMERA	JST-SH 1.0 mm 3P
CAN	JST-GH 1.25 mm 4P
DJI O3	JST-SH 1.0 mm 6P
PWM pads	outer column · silk S5–S10
ESC pads	inner column · same M1–M4 as the ESC port
Edge pads	SPI3 · UART3 · buzzer
Top pads	UART1 · WS2812 LED · I2C2
Bottom pads	UART2 · UART8 · CAN1 · 12 V
Camera power	jumper JP1 · cut 5V side, bridge VBAT

2. Specifications

· Hardware Specifications (1)

MCU	STM32H743VIH6, 480 MHz (1 MB RAM, 2 MB Flash)
IMU	2× ICM-42688-P
Barometer	DPS368
Compass	IST8310
OSD	AT7456E analog OSD camera in, VTX out
Operating Voltage	2–8S LiPo (up to 33.6 V) · both regulators rated 60 V, 36 V TVS on the battery input
Output Voltage	5 V / 2 A · 12 V / 2 A (VTX, DJI O3) · 5 V port rail for GPS, receiver and CAN that also runs from USB alone
ESC Port	10-pin JST-SH: GND · GND · VBAT · VBAT · M1–M4 · CUR · TEL straight 1:1 cable to the novaX AE-8S-60A 4-in-1 V2 ESC
Current & Telemetry	ESC current on the ESC port (PC1, 40 A per volt, same scale as the ESC) · ESC telemetry on SERIAL7 (UART7 RX)

2. Specifications

· Hardware Specifications (2)

PWM Output	10 motor/servo channels (M1–M4 on the ESC port and pads, S5–S10 on pads) + 1 WS2812 LED pad
Motor Order	Betaflight order out of the box 1 rear right · 2 front right · 3 rear left · 4 front left; ArduPilot default FRAME_TYPE 12, so the 1:1 ESC cable needs no remapping
Power Select Pads	Camera 5 V or VBAT · VTX 12 V or VBAT 3-pad solder jumpers on the bottom; VBAT only for devices rated for the pack
Buzzer / Switch	BZ+ pad (5 V or 12 V by solder jumper) and BZ– on a 30 V 5.8 A MOSFET a buzzer, or on/off for another device
Production Pads	VBUS · D+ · D– · GND · BOOT0 · 3V3 test pads on a 2.54 mm row flashing jigs program several boards over one USB hub, no button
Ports	GPS + compass · ELRS/RC · analog VTX · analog camera · DJI O3 · OpenIPC · CAN · SWD pads

2. Specifications

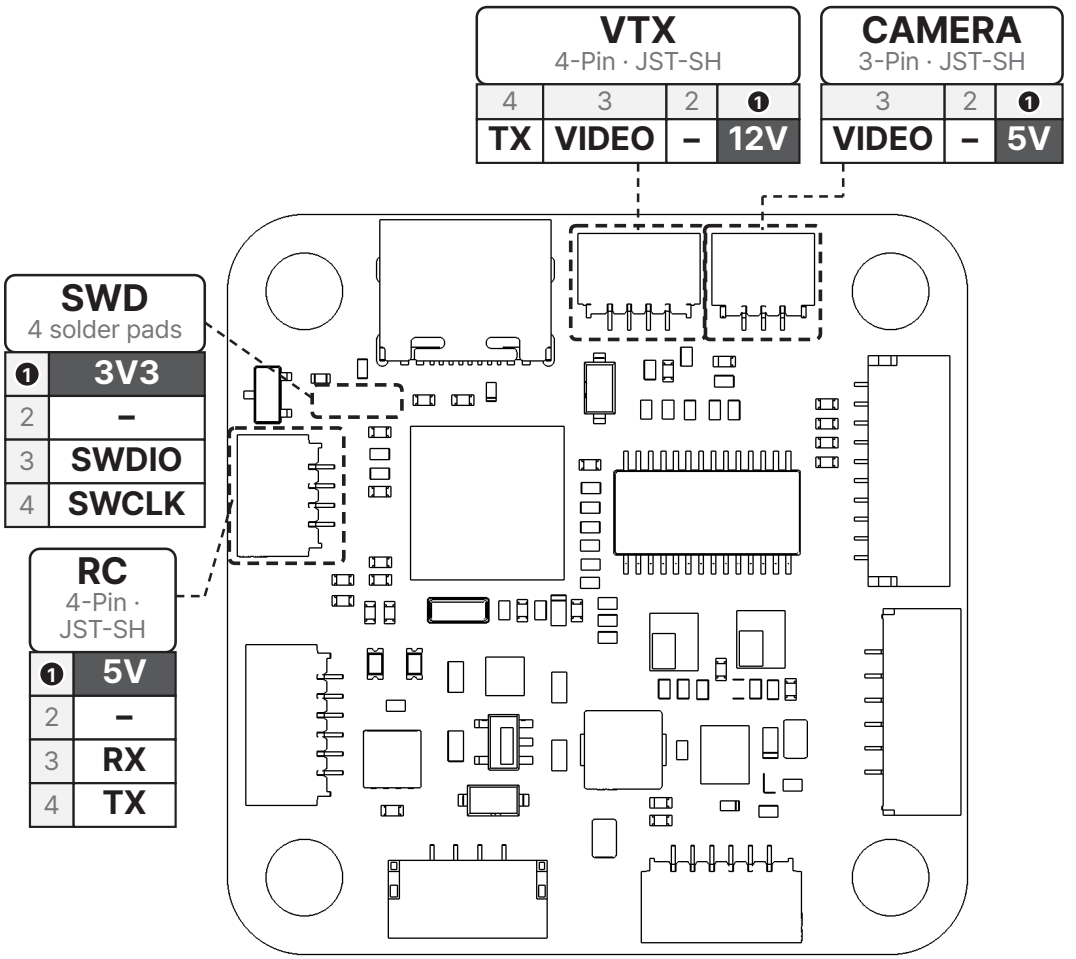
· Hardware Specifications (3)

CAN	1 port · TI TCAN3413 3.3 V CAN FD transceiver works on USB power
USB Input	USB-C · 5.1 kΩ CC resistors, so C-to-C cables and chargers power it
Blackbox	microSD card
Size	38.5 × 38.5 mm
Mount Hole	30.5 × 30.5 mm / Ø4 mm
Core Chips	MCU ST · IMU TDK · barometer Infineon · regulators MPS · LDOs and CAN TI non-Chinese makers; the analog OSD chip is the exception
Validation Status	Design stage schematic and part placement done and checked (all nets pin for pin, 3 identical passes); routing and samples next. No firmware release yet.

3. Pinouts

· Top (1)

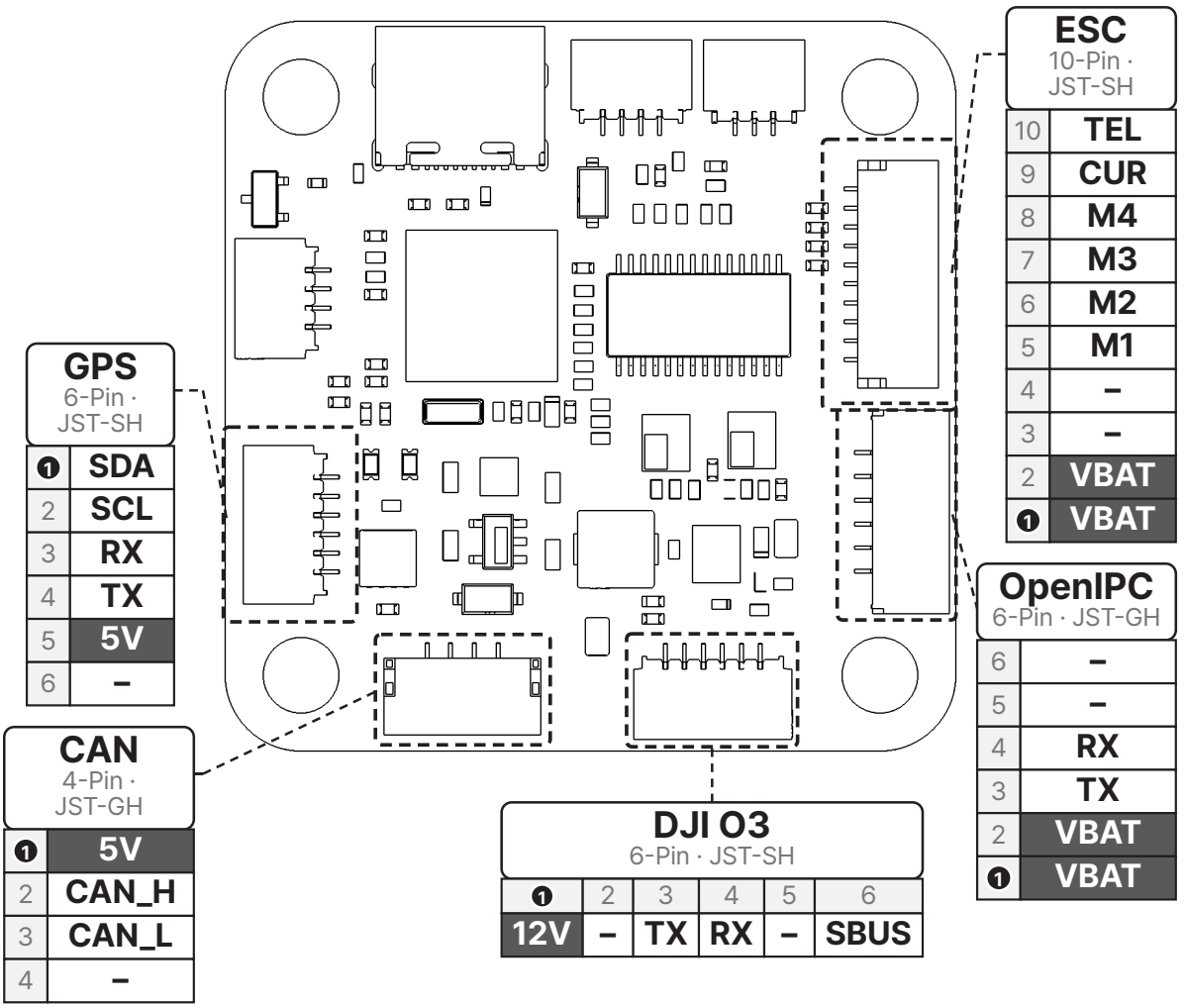
❶ = pin 1 **5V** = power **–** = GND **TX** = signal · pins as seen in the picture



3. Pinouts

· Top (2)

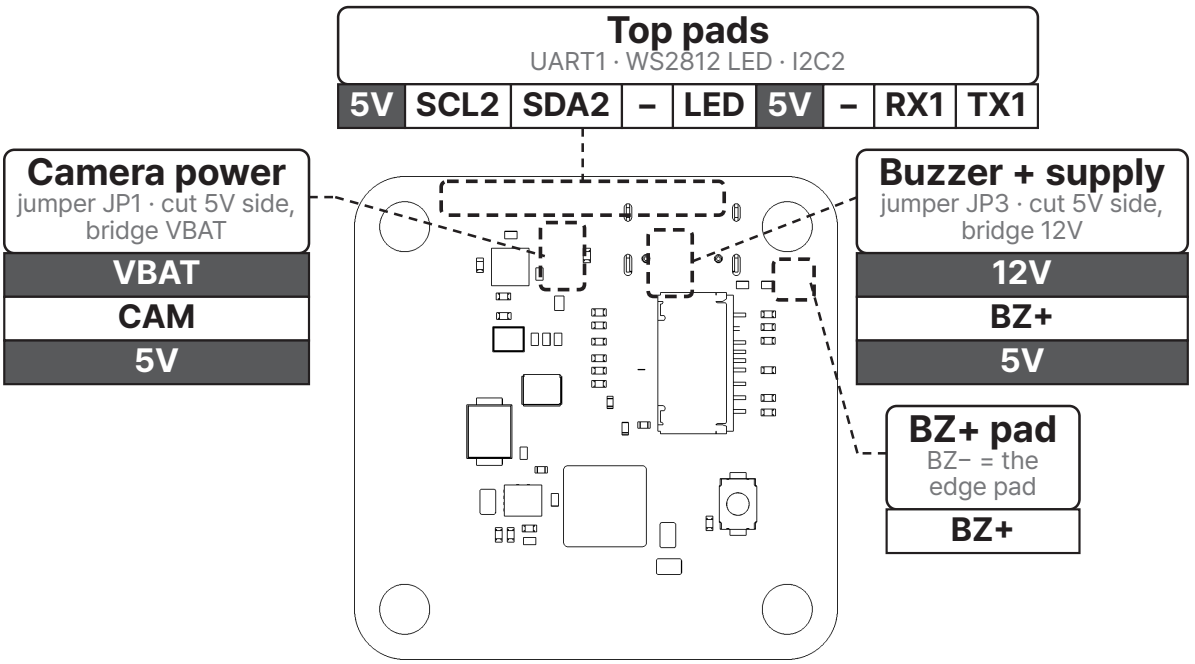
❶ = pin 1 **5V** = power **–** = GND **TX** = signal · pins as seen in the picture



3. Pinouts

· Bottom (1)

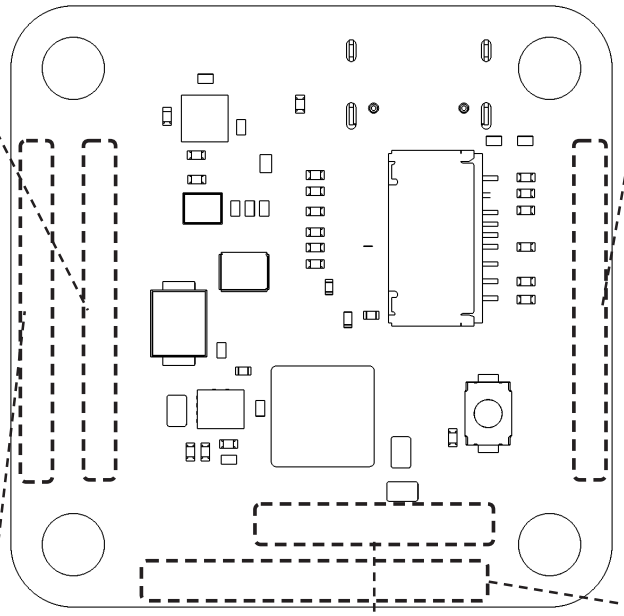
❶ = pin 1 **5V** = power **–** = GND **TX** = signal · pins as seen in the picture



3. Pinouts

· Bottom (2)

❶ = pin 1 **5V** = power **–** = GND **TX** = signal · pins as seen in the picture



ESC pads
inner column · same M1–M4 as the ESC port

M1
M2
M3
M4
–
VBAT
RX4
TX4

PWM pads
outer column · silk S5–S10

S5
S6
S7
S8
S9
S10
5V
–

Edge pads
SPI3 · UART3 · buzzer

BZ–
SCK
MISO
MOSI
CS2
CS1
RX3
TX3

Bottom pads
UART2 · UART8 · CAN1 · 12 V

RX2
TX2
12V
RX8
TX8
5V
CANL
CANH
–

Flashing jig pads
D1.0 mm · 2.54 mm · BOOT0 to 3V3 = USB DFU

3V3	BOOT0	–	D–	D+	VBUS

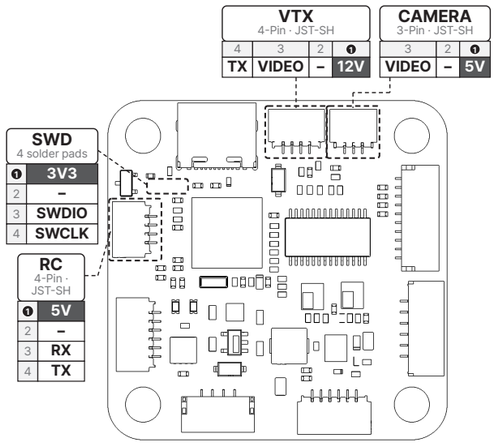
novaX

AF-H7 nano V2

3. Pinouts

· Top (1)

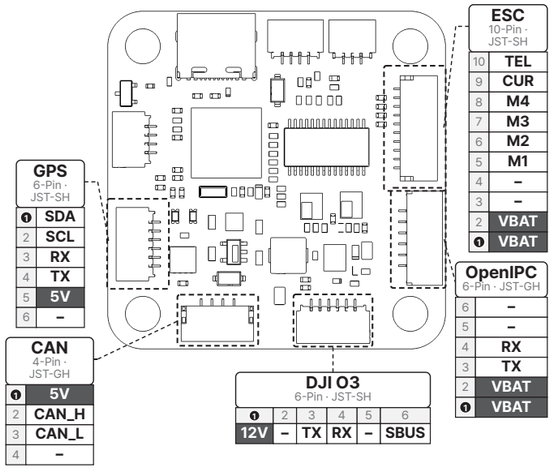
❶ = pin 1 · 5V = power · – = GND · TX = signal · pins as seen in the picture



3. Pinouts

· Top (2)

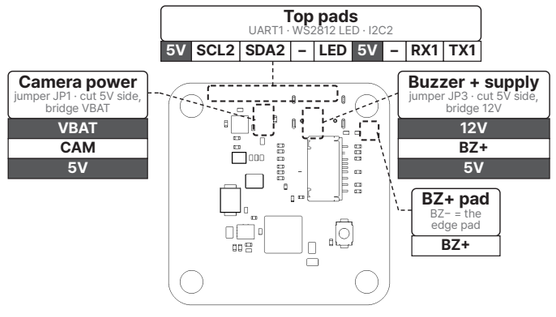
❶ = pin 1 · 5V = power · – = GND · TX = signal · pins as seen in the picture



3. Pinouts

· Bottom (1)

❶ = pin 1 · 5V = power · – = GND · TX = signal · pins as seen in the picture



3. Pinouts

· Bottom (2)

❶ = pin 1 · 5V = power · – = GND · TX = signal · pins as seen in the picture

